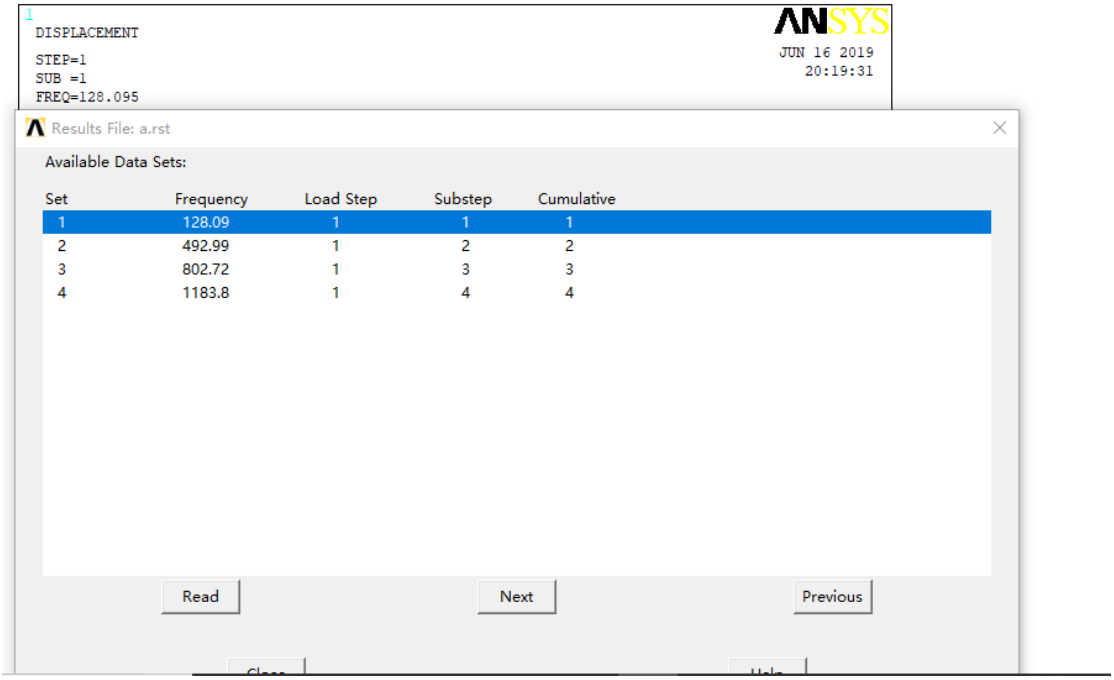
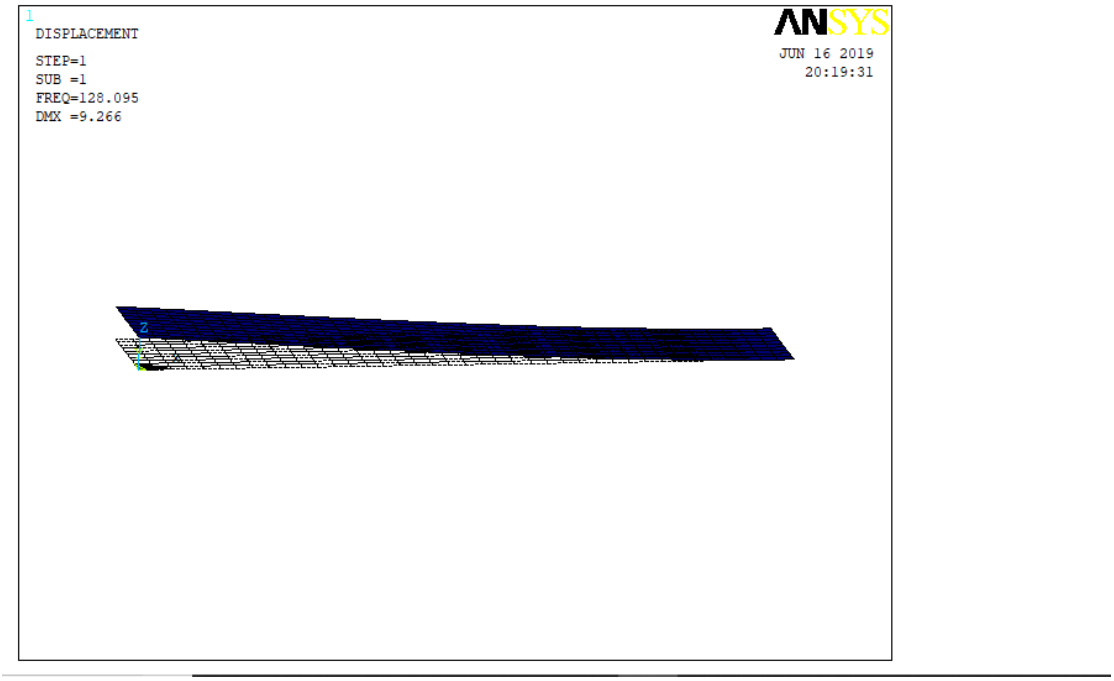
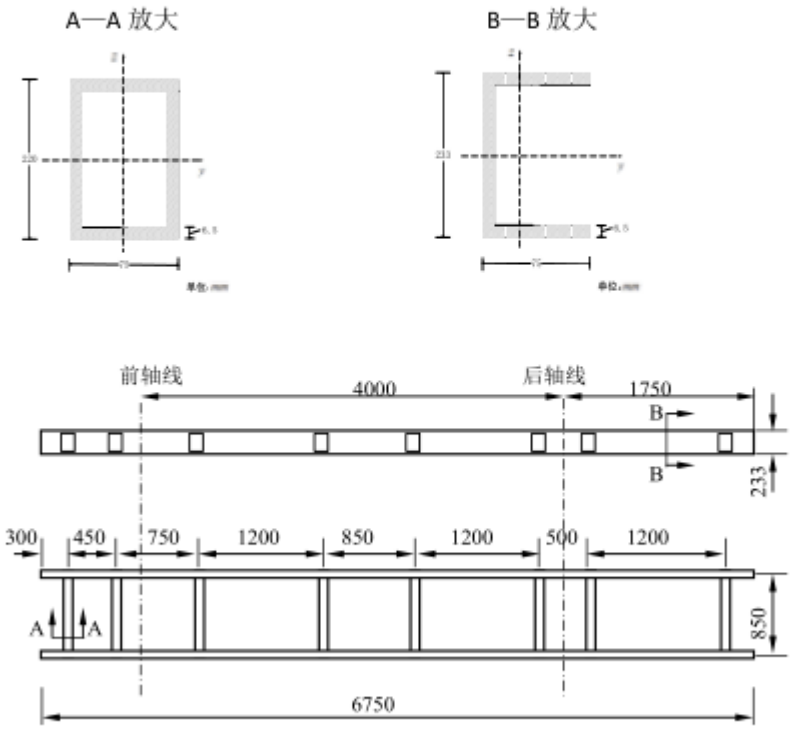
例题 1


1
DISPLACEMENT
STEP=1
SUB =1
FREQ=128.095
ANSYS
JUN 16 2019
20:19:31
Results File: a.rst
Available Data Sets:
Set Frequency Load Step Substep Cumulative
1 128.09 1 1 1
2 492.99 1 2 2
3 802.72 1 3 3
4 1183.8 1 4 4
Read
Next
Previous


1
DISPLACEMENT
STEP=1
SUB =1
FREQ=128.095
DMX =9.266
ANSYS
JUN 16 2019
20:19:31

上机实践作业（三）：

图示为汽车车架结构简图。车架纵梁为槽形钢(开口向内，且左右纵梁横截面形心之间的间距为 850mm)，横梁均为矩形管钢，左右对称，其尺寸见图。车架材料的弹性模量为 200GPa，泊松比为 0.3，密度为 7840kg/m^3。试用有限元法对车架进行固有振动分析（要求计算前 10 阶固有振动频率和振型）



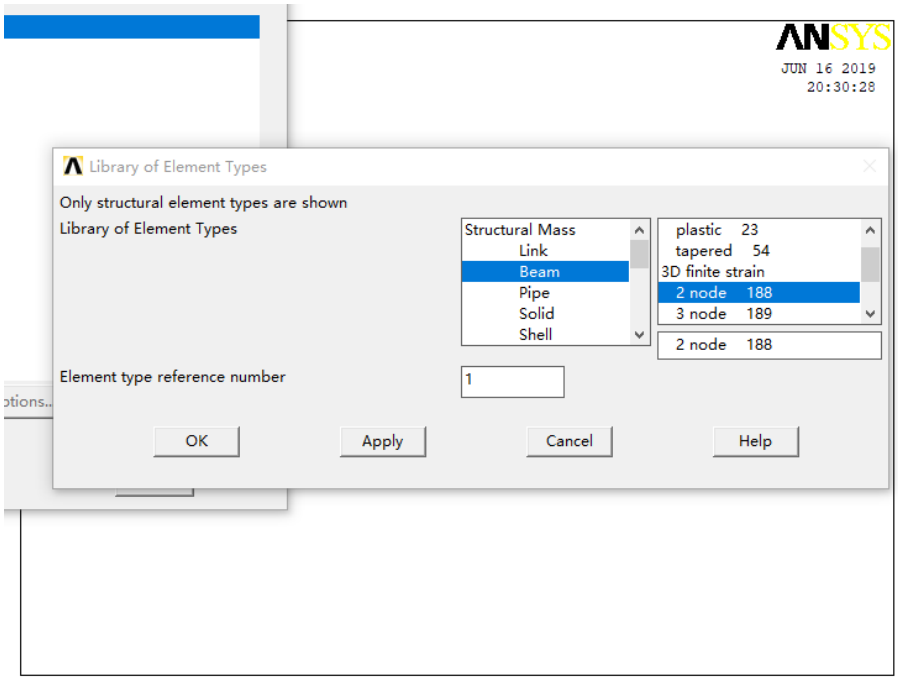
单位：mm；刚板厚 6.5mm

车架结构简图

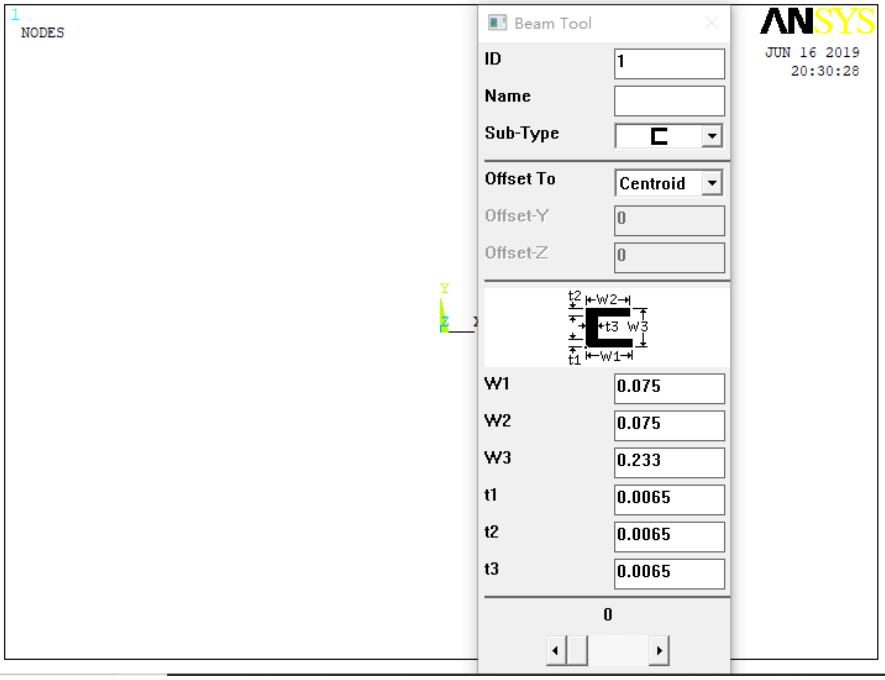
1) 点击开始，ANSYS ED 10.0 ——ANSYS Product Launcher 点击 File Management ，在 Working directory 处改变工作目录，，在 Initial Jobname 中输入文件名，点击 Run。

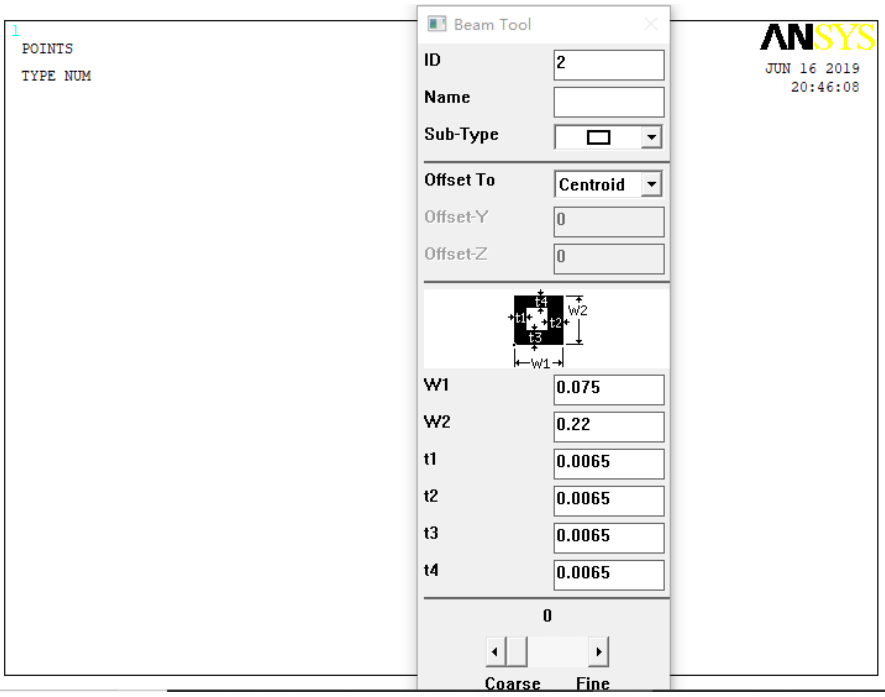
2) 点击 Preference——选择 Structural，OK。

3) 定义单元类型。ANSYS Main Menu: Preprocessor—Element Type—Add/Edit/Delete —Add，选择 2 node 188 单元，OK。点击 Close 关闭单元定义窗口。

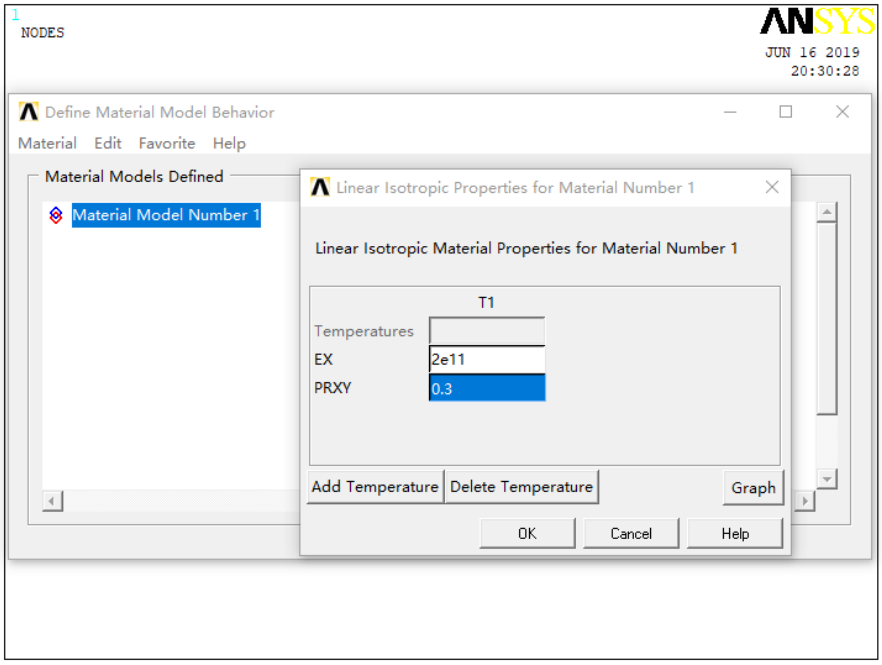


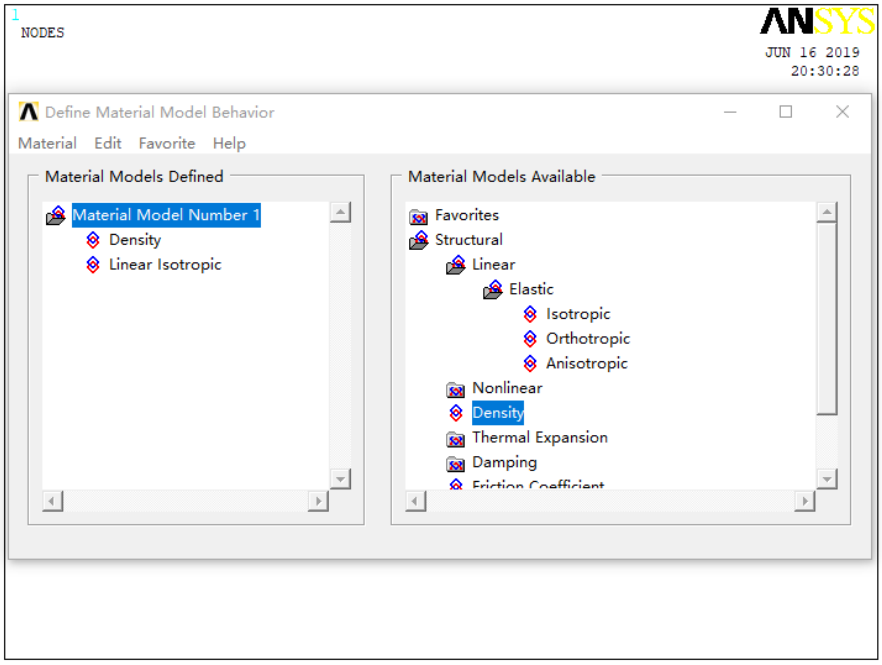
4) 定义单元截面参数。ANSYS Main Menu:Preprocessor— Section— Beam— Common Sections，在 Sub-Type 中选择槽钢截面，ID 为 1，设置如下第一幅图所示参数，点击 OK；重复操作选择矩形管截面，ID 为 2，设置如下第二幅图所示参数,点击 OK。



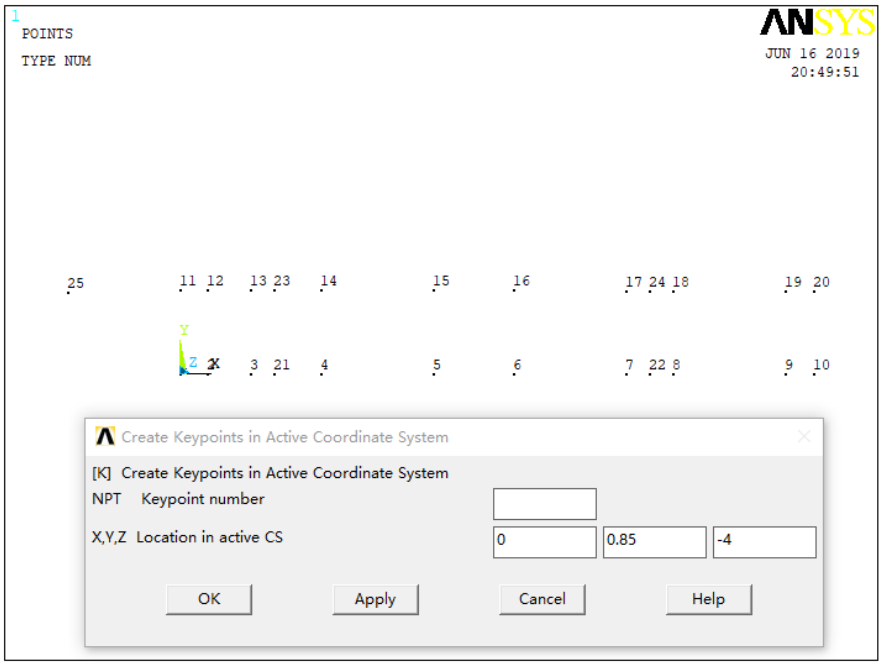


5) 定义单元材料属性。ANSYS Main Menu:Preprocessor——Material Props——Material Models——Structural——Linear——Elastic——Isotropic——输入 E=2e11，μ=0.3，OK。同时在 Structural——Density 输入密度 DENS=7840，关闭材料定义窗口。

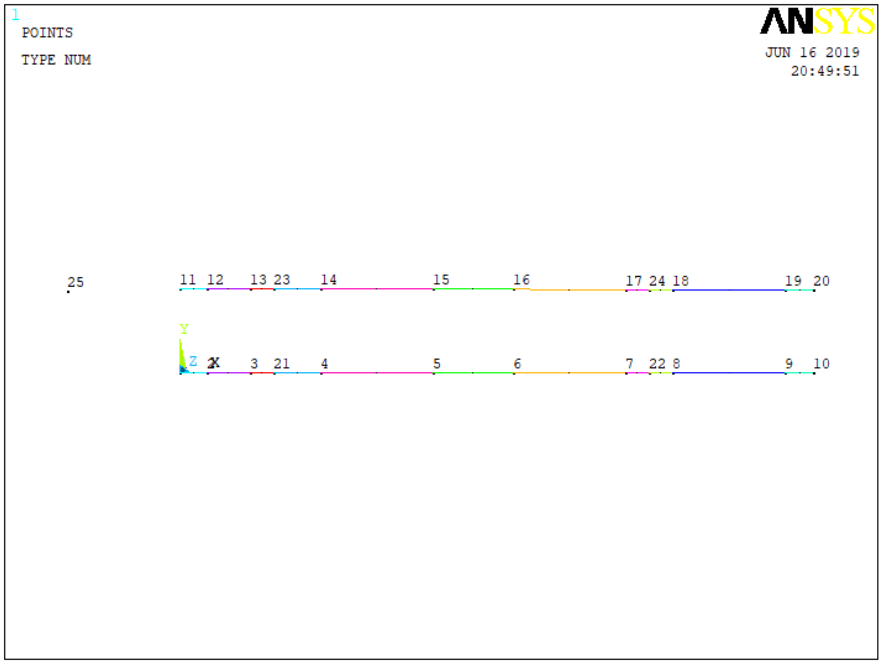




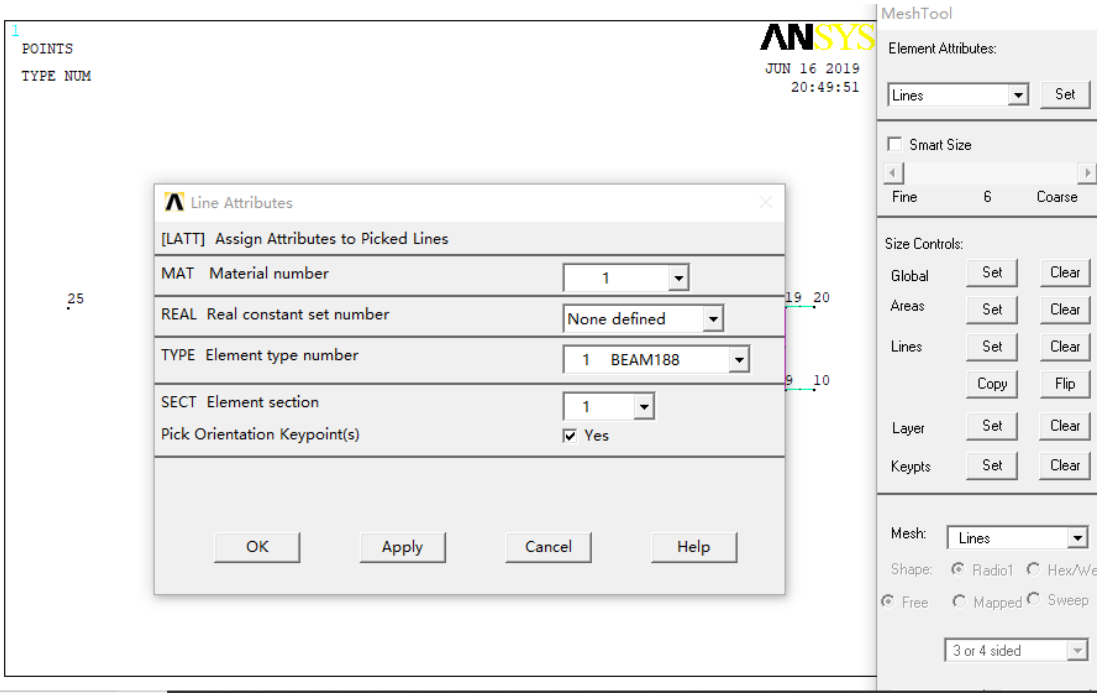
6) 创建关键点。采用 ANSYS Main Menu: Preprocessor— Modeling— Create— Keypoints —In Active CS 输入 相应车架结构相应 24 个点坐标，然后输入点（0，0.85，-4），点击 OK。



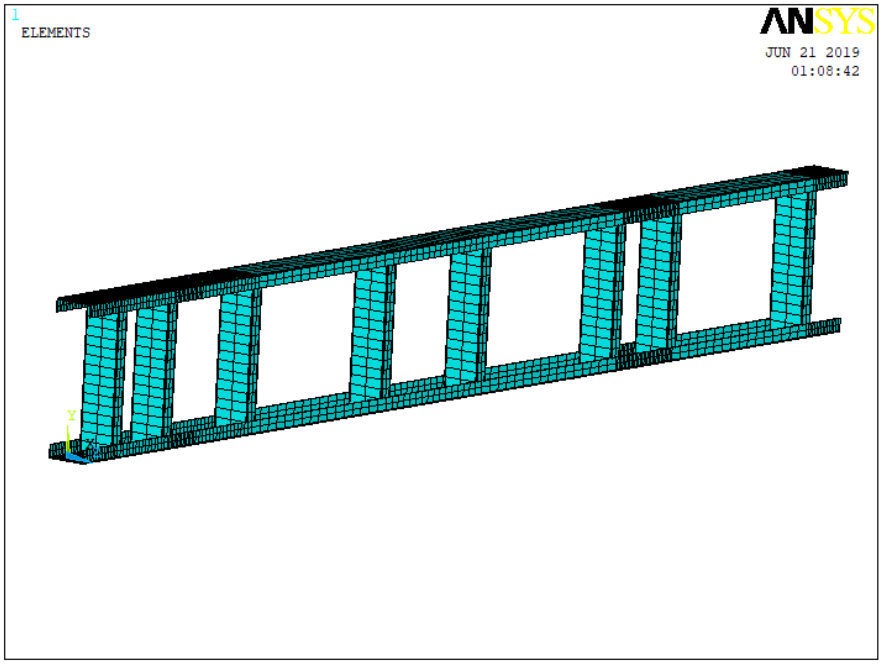
7) ANSYS Main Menu: Preprocessor— Modeling— Create— Lines— Lines—Straight Line，依次相应点，建立线段，点击 OK。



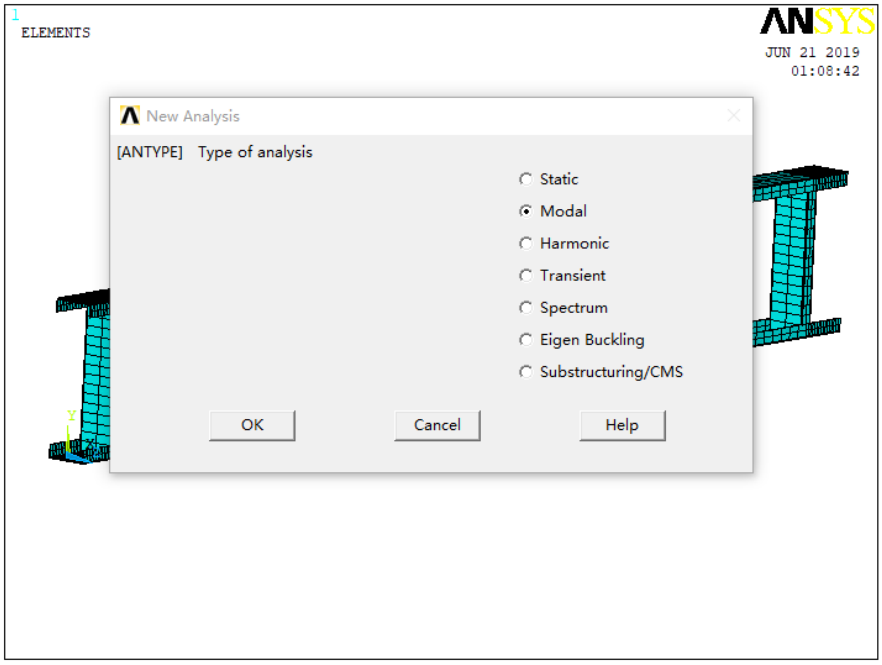
8) 划分网格。ANSYS Main Menu: Preprocessor — Meshing- MeshTool，在上方 ElementAttributes 中选取 Lines，点击右侧 Set，选取下侧纵梁，点击 OK，在 SECT Elementsection 中选取 1，点击 OK；重复操作，选取剩余纵梁，点击 OK，在 SECT Element section 中选取 1，pick orientation keypoint 点上“√”号，点击 OK，然后选取 K 点，点击 OK；重复操作，选取所有横梁，在 SECT Element section 中，选取 2，pick orientation keypoint 去掉“√”号点击 OK。



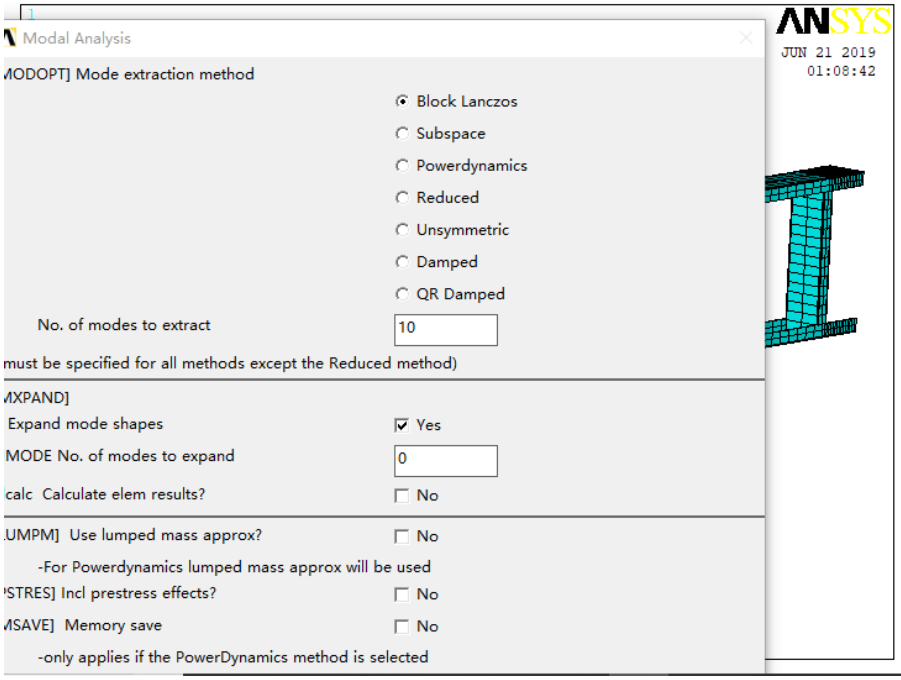
Preprocessor- Meshing- MeshTool,选择 Size Controls 下 Lines，右侧的 set 设置单元大小，Pick All，在 NDIV No. of element divisions 右侧输入 16;Preprocessor- Meshing- MeshTool，点击 Mesh，点击 Pick All。

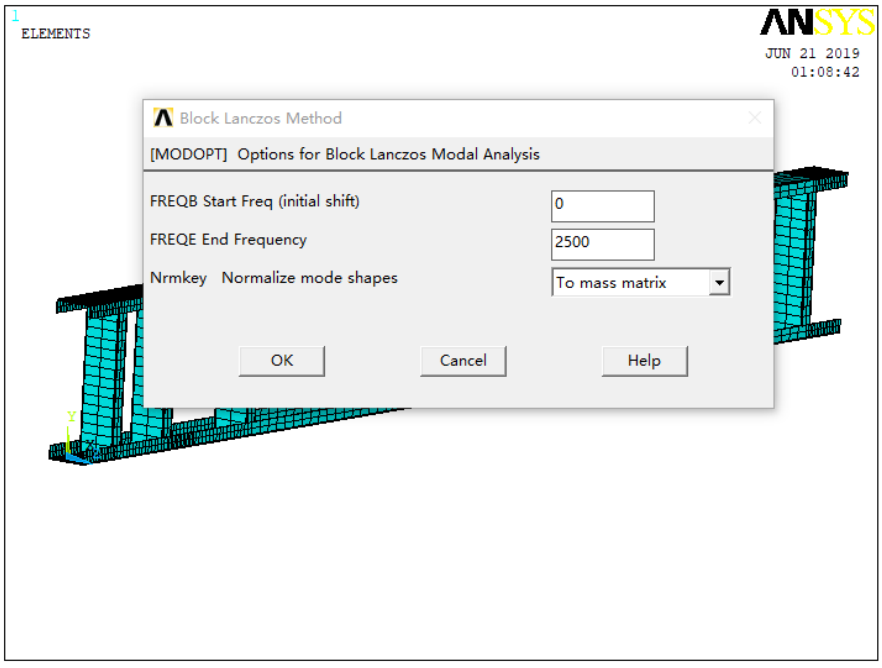


9) 设置分析类型。Solution——Analysis Type——New Analysis——Modal。

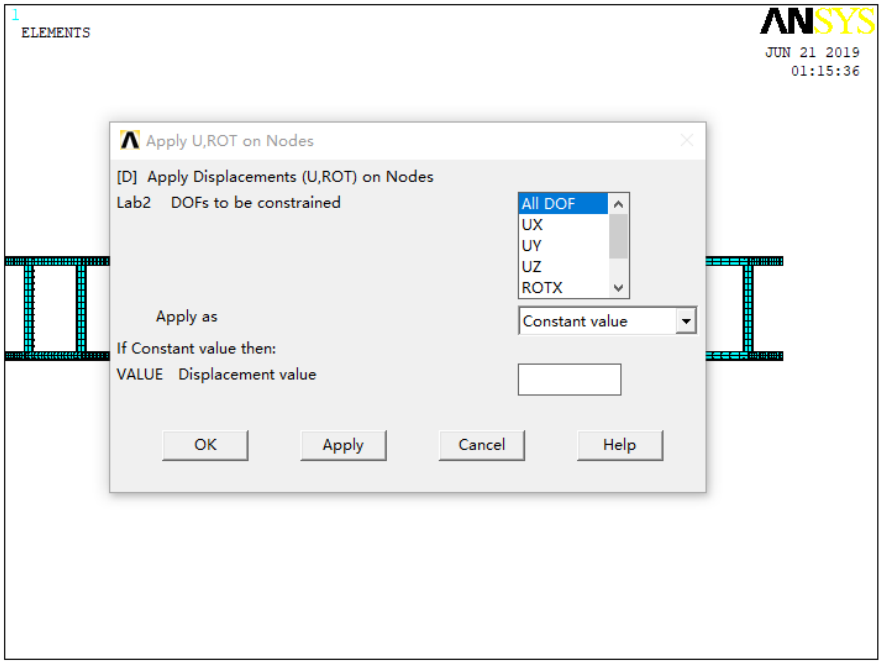


10) 设置提取的模态阶数及范围。Solution——Analysis Type——Analysis Option——Block Lanczos——No. of modes to extract：10——OK——Start Freq：0；End Freq：2500——OK。



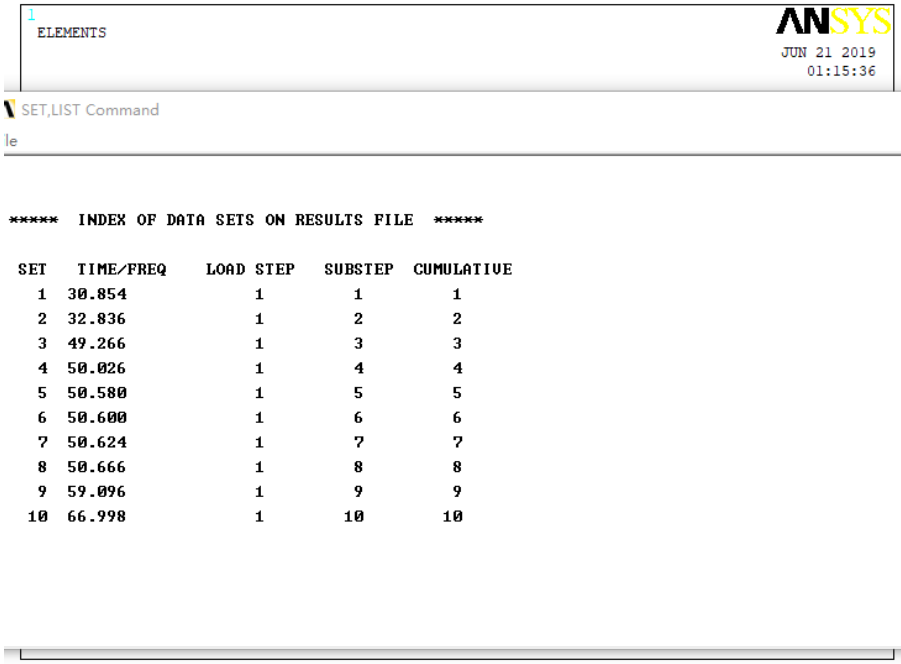


11) 施加约束。ANSYS Main Menu: Preprocessor- Define Loads- Apply- Structural-Displacement-On Nodes，选择四个支点，OK，选取 ALL DOF。

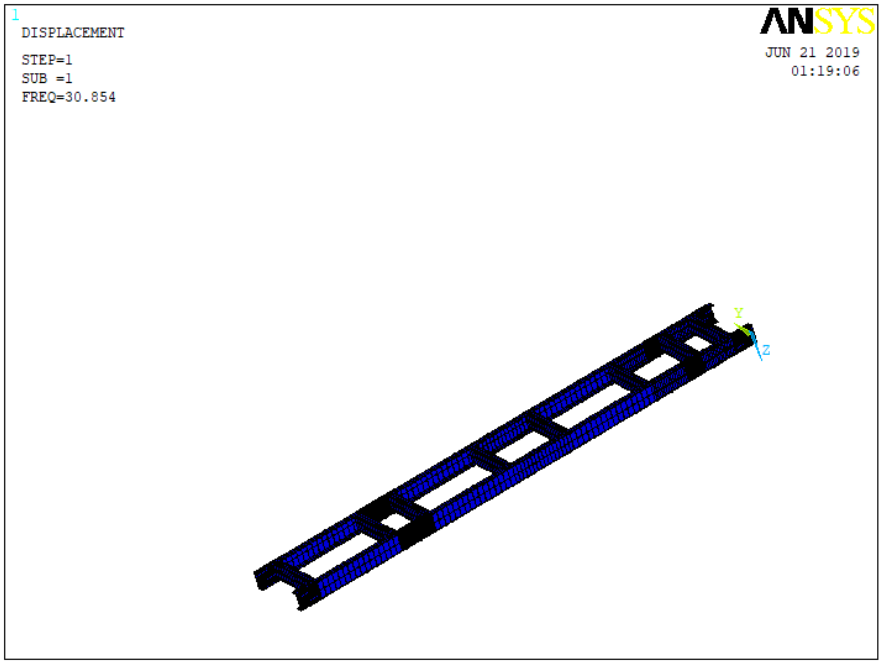


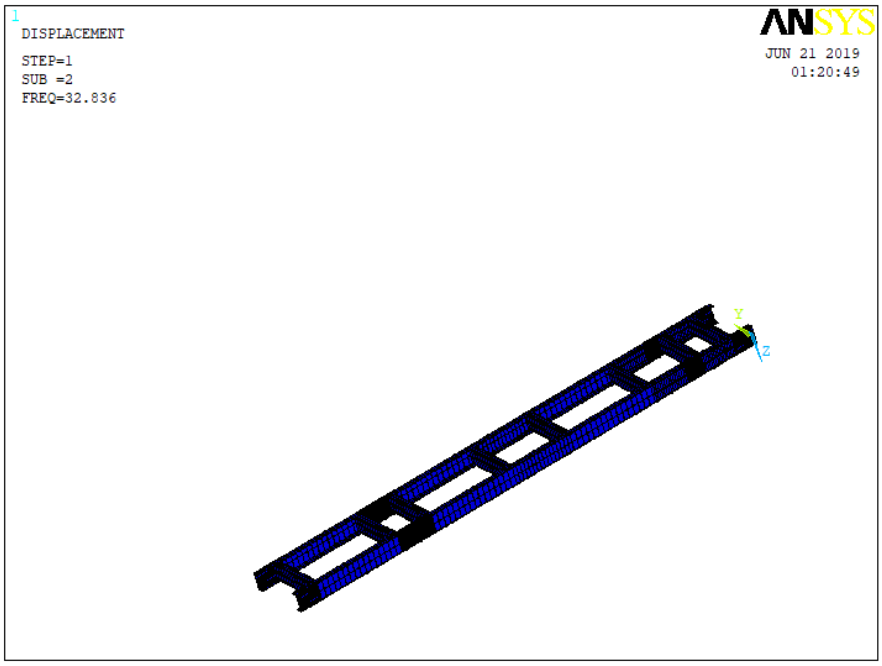
12) Solution——Solve——Current LS，OK。

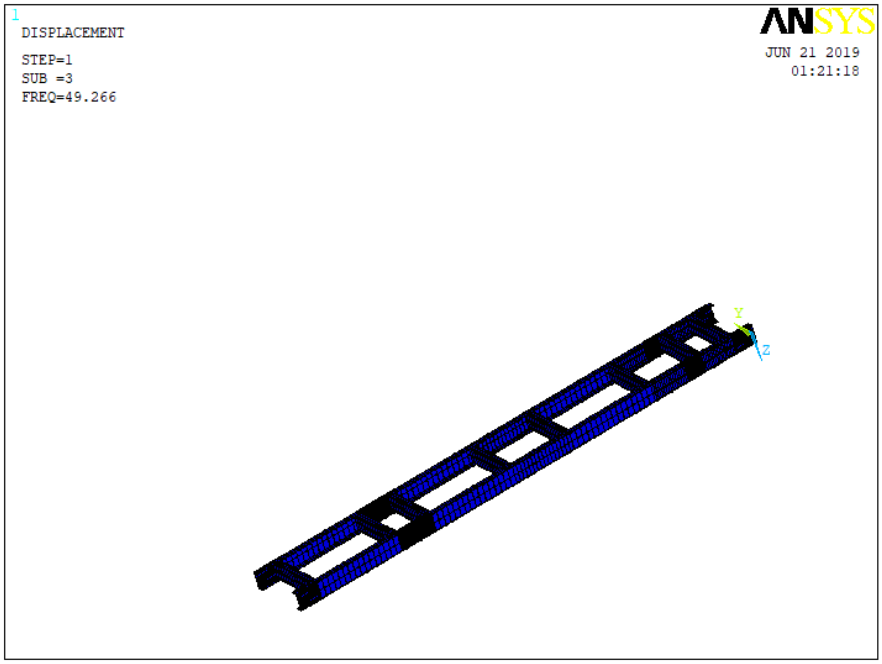
13) 查看计算出的固有频率值。ANSYS Main Menu： ： General Postproc——Results Summery。

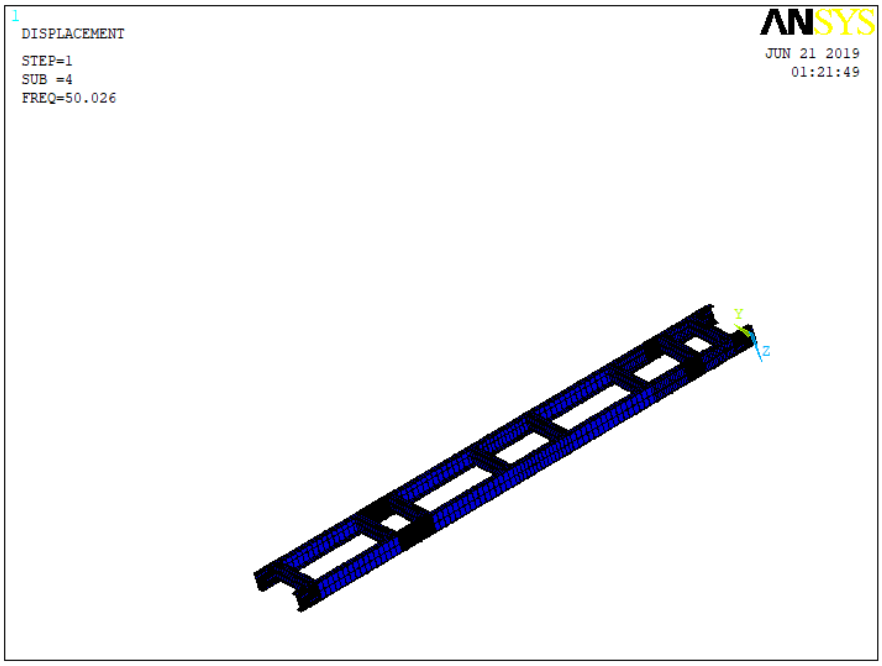


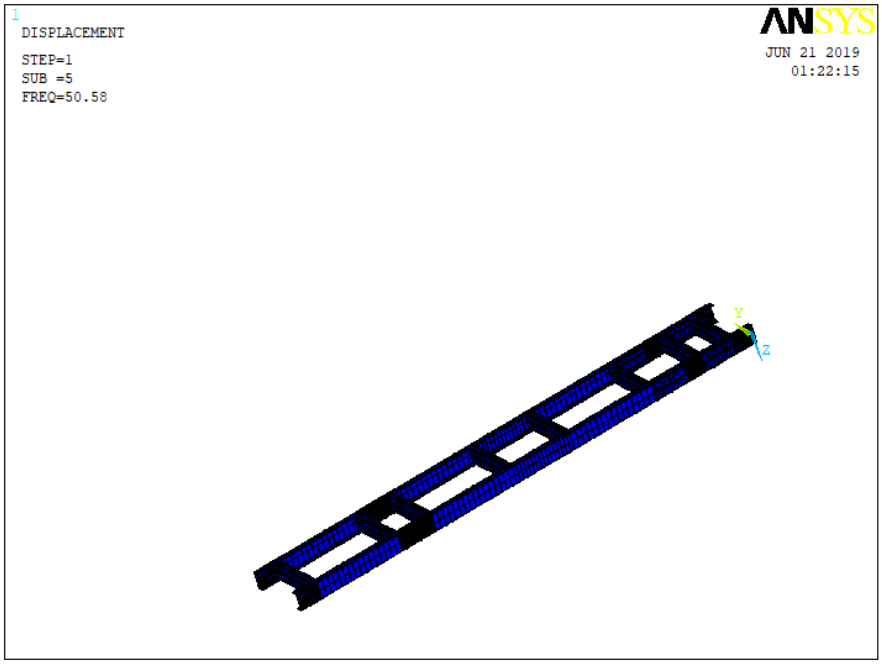
14）依次查看全部 10 阶模态振型结果。General Postproc——read results——By Pick 依次选择一个结果，read。关闭窗口。 General Postproc——Plot Results——Deformed Shape，可以看到前面选择的振型的结果。或者 Utility Menu——PlotCtrls——Animate—— Mode Shape 中查看动画。

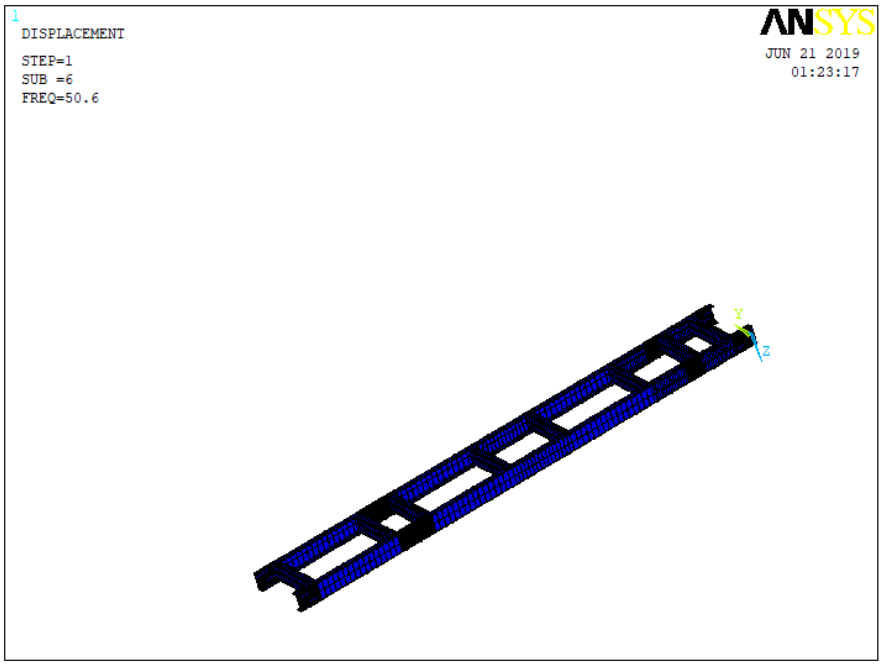

1
DISPLACEMENT
STEP=1
SUB =1
FREQ=30.854
ANSYS
JUN 21 2019
01:19:06
Y
Z

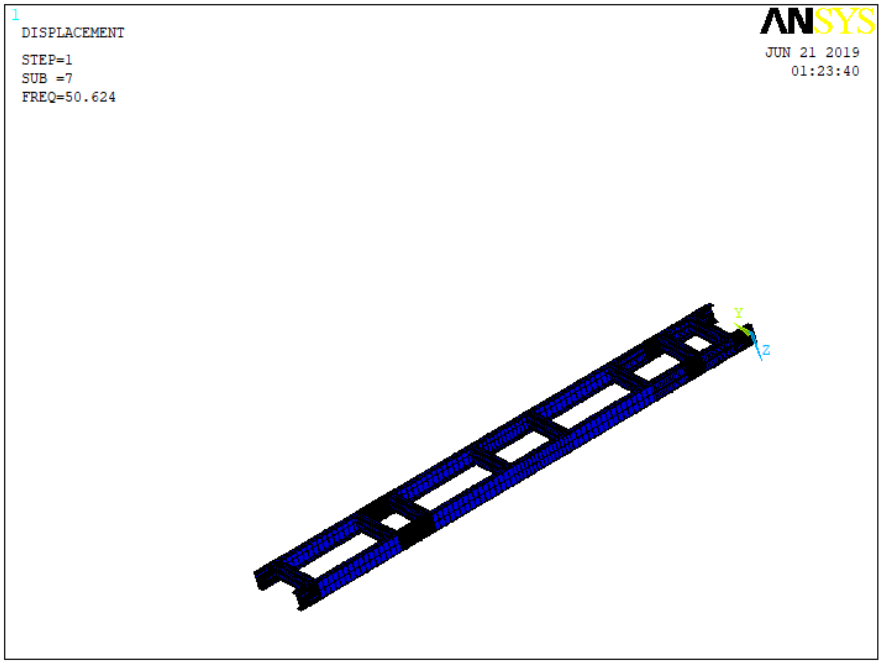

1
DISPLACEMENT
STEP=1
SUB =2
FREQ=32.836
ANSYS
JUN 21 2019
01:20:49
Y
Z

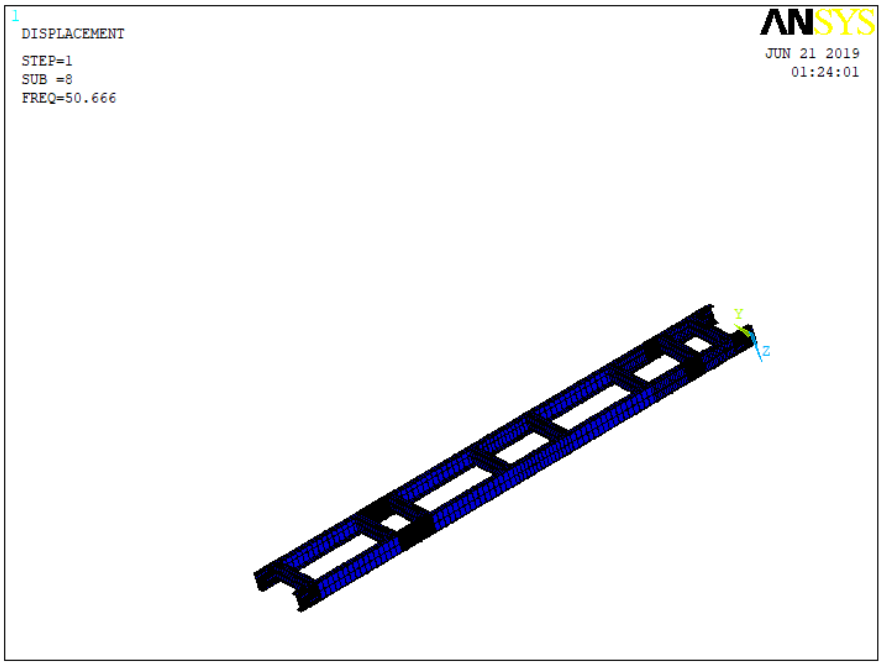

1
DISPLACEMENT
STEP=1
SUB =3
FREQ=49.266
ANSYS
JUN 21 2019
01:21:18
Y
Z

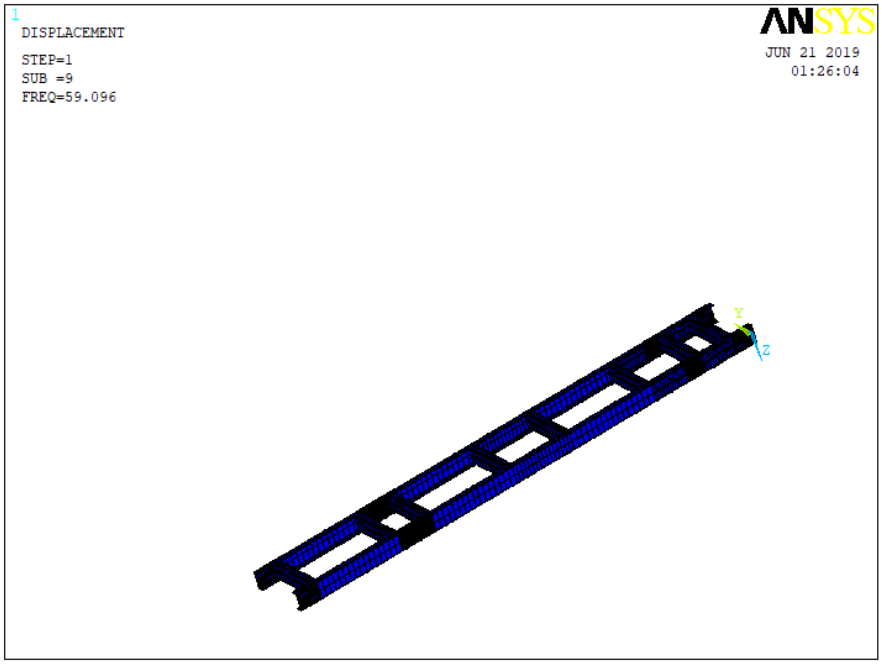

1
DISPLACEMENT
STEP=1
SUB =4
FREQ=50.026
ANSYS
JUN 21 2019
01:21:49
Y
Z

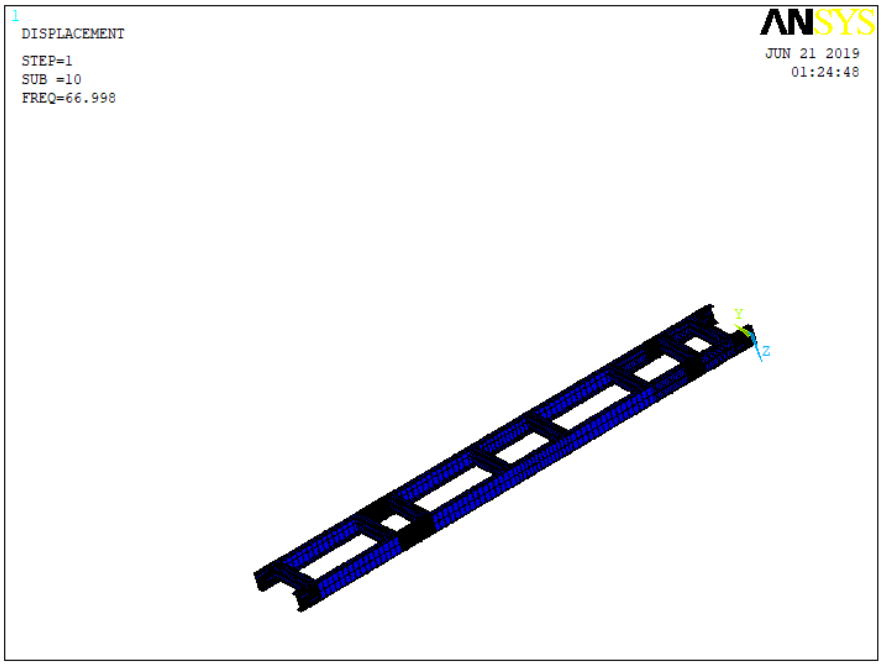

1
DISPLACEMENT
STEP=1
SUB =5
FREQ=50.58
ANSYS
JUN 21 2019
01:22:15
Y
Z


1
DISPLACEMENT
STEP=1
SUB =6
FREQ=50.6
ANSYS
JUN 21 2019
01:23:17
Y
Z


1
DISPLACEMENT
STEP=1
SUB =7
FREQ=50.624
ANSYS
JUN 21 2019
01:23:40
Y
Z


1
DISPLACEMENT
STEP=1
SUB =8
FREQ=50.666
ANSYS
JUN 21 2019
01:24:01
Y
Z


1
DISPLACEMENT
STEP=1
SUB =9
FREQ=59.096
ANSYS
JUN 21 2019
01:26:04
Y
Z


1
DISPLACEMENT
STEP=1
SUB =10
FREQ=66.998
ANSYS
JUN 21 2019
01:24:48
Y
Z